Earthing system (IEC 364)	
TNS	☒
TNC	☐
IT	☐
TT	☐

Carpentry
Protection degree IP42 (EN60259,IEC60259)
Ral 7035 inside & outside

Refrence standards IEC60439-1

Wire sizes (PVC Wires) Inside of panel at 40°C	
Line/ Feeder Current	Wire size
Control <= 8A	1.0 mm²
11A	1.5 mm²
18A	2.5 mm²
23A	4 mm²
30A	6 mm²
44A	10 mm²
58A	16 mm²
77A	25 mm²
105A	35 mm²
142A	50 mm²
169A	70 mm²
212A	95 mm²
240A	120 mm²

Protection method against direct contacts (IEC60439-1)	
Method A	☒

Cable/wire color codes

Circuits	Rated Voltage	Cable/wire color	Cable/wire Type
AC power cables	230-440VAC	Black	PVC/PVC,NYY
DC Control cables	24VDC	Grey	PVC/PVC NYY Shielded
AC Power Circuit wires	230-440VAC	Black	PVC/PVC,NYY
Neutral	230-440VAC	Light Blue	PVC,NYY
DC Auxiliary Circuit wires L+	24VDC	Red	PVC,NYY
DC Auxiliary Circuit wires 0VDC	0VDC	Dark Blue	PVC,NYY
Clean Earth (IE)		Green-Yellow with Blue Ring	PVC,NYY
Externally supplied circuits	230-440VAC	Orange	PVC,NYY
Digital Signals	24VDC	Grey	PVC,NYY
Analogue Signals	24VDC	Brown	PVC,NYY

Main circuit identification

Alternating current	Direct current	Earthing system
Phase1 L1	Positive L+	Protection circuit PE
Phase2 L2	Negaitive L-	
Phase3 L3	Neutral 0V	
Neutral N		

REFERENCE

REFERENCE. TITLE	REFERENCE. No.

NOTES

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04	8/25/2020	ISSUED FOR CONSTRUCTION	M.A	M.G	K.Z	
03	5/9/2020	ISSUED FOR CONSTRUCTION	M.A	M.G	K.Z	
02	4/8/2020	ISSUED FOR APPROVAL	M.A	M.G	K.Z	
01	3/2/2020	ISSUED FOR APPROVAL	M.A	M.G	K.Z	
00	2/10/2020	ISSUED FOR REVIEW	M.A	M.G	K.Z	

Client: NATIONAL IRANIAN COPPER INDUSTRIES COMPANY (NICICO)

PMC: PARMAN POOYESH CO.

Consultant: MEMSECO & ARIA HANGARD JOINT VENTURE

Contractor: CANY MES CO.

Project: SUNGUN Copper Ore Heap Leaching Plant Project

Title: Electrical Schematic Wiring Diagrams for RO Package
Schematic Wiring
Standards

Document Number:	DATE	SIZE	SCALE	SHEET NO.	REV. CLIENT	REV. VENDOR
11L1-AV-P-220-EL-DIG-1210	8/25/2020	A3	NA	6 OF 35	04	04

Electrical Symbol	Description	Electrical Symbol	Description	Electrical Symbol	Description
MCCB 630A 500-630A	3P moulded case circuit breaker	16A	2P Miniature circuit-breaker	-PL1 20W	Panel lighting
MPCB 30-40A	3P Motor protection Circuit Breaker	-FAN1	Panel fan	-K	Relay/contactor coil
6AgG	3P switchfuse	-E1 230 V	Panel heater	VFD 4KW	VARIABLE FREQUENCY DRIVE (VFD)
600/5A	3 Nr. current transformer	Phase Seq. c.	Phase sequence Control	-V1 220/24VDC,10A DRP240-24	Power supply 220VAC/24VDC/10A
Contactor 18.5KW	3P Contactor Single line	-RX Red	Signal Lamp	-SO1	Din rail mounted socket 1P,16A,230VAC
Multi Meter	Digital multimeter	-SAX L O R LOCAL REMOTE	Remote-off-Local Selector switch	3KW 6A	3P Asynchronous Motor
	Withdrawable Power terminal	-XCX:2	Withdrawable control terminal	SSX SERVICE	Service/Test switch

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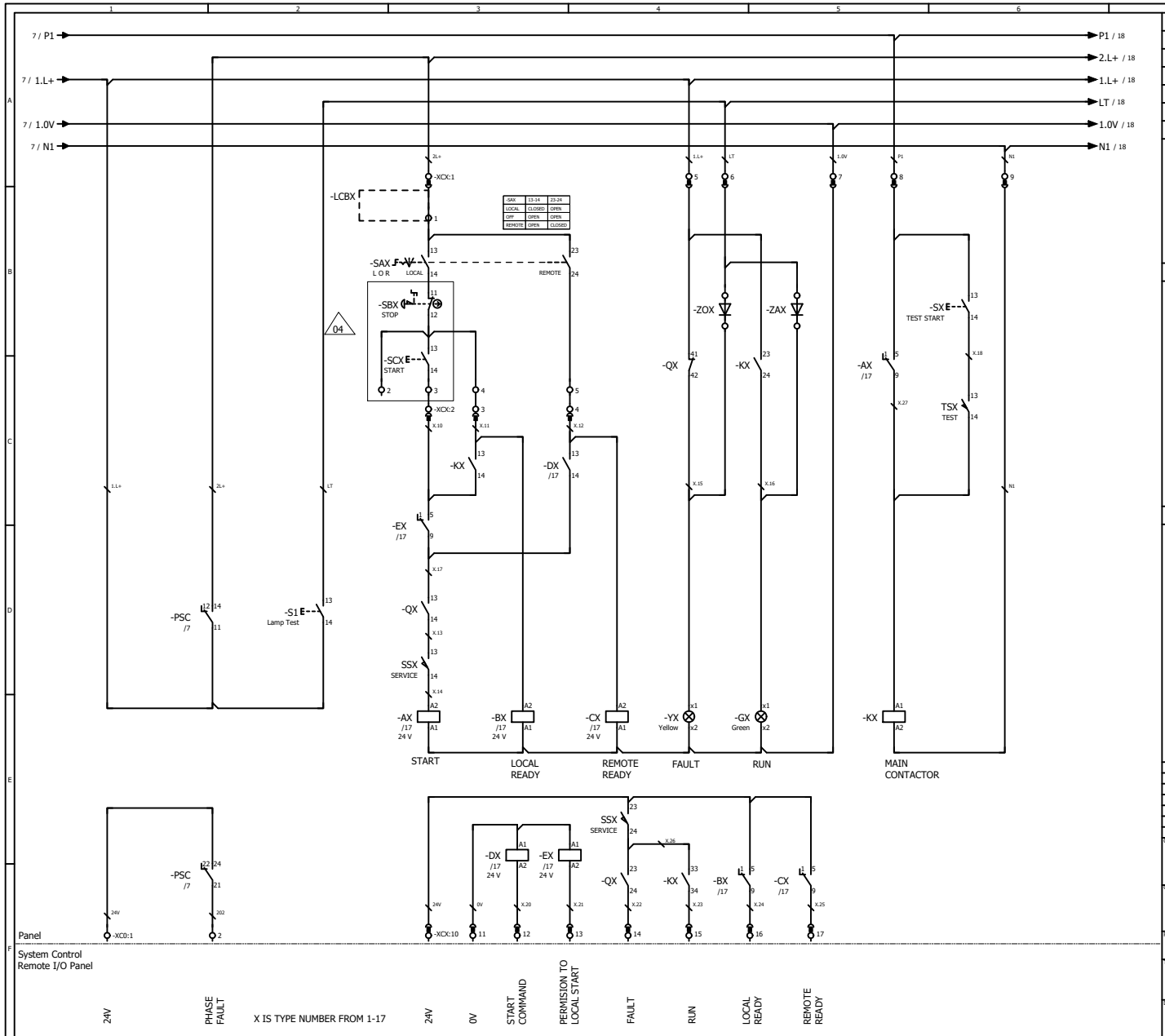
Legend

Document Number: 11L1-AV-P-220-EL-DIG-1210

DATE	SIZE	SCALE	SHEET NO.	REV. CLIENT	REV. VENDOR
8/25/2020	A3	NA	4 OF 35	04	04



-SAX	13-14	23-24
LOCAL	CLOSED	OPEN
OFF	OPEN	OPEN
REMOTE	OPEN	CLOSED

REFERENCE

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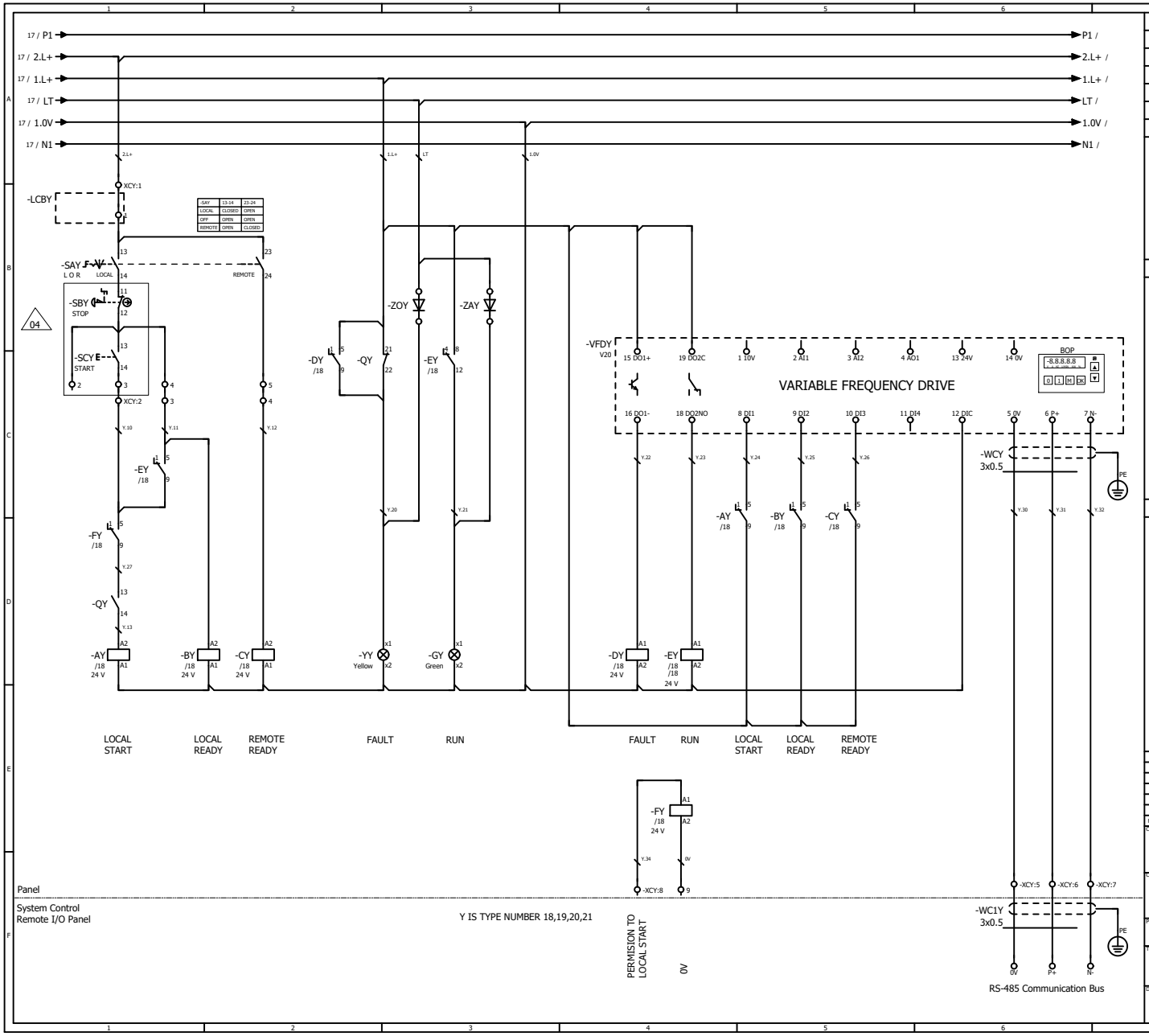
Schematic Wiring
TYP_DOL_Control

Document Number:	DATE	SIZE	SCALE	SHEET NO.	REV. CLIENT	REV. VENDOR
11L1-AV-P-220-EL-DIG-1210	8/25/2020	A3	NA	17 OF 35	04	04



-SAY	13-14	23-24
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OFF	OPEN	OPEN
REMOTE	OPEN	CLOSED

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TYP_VFD_Control

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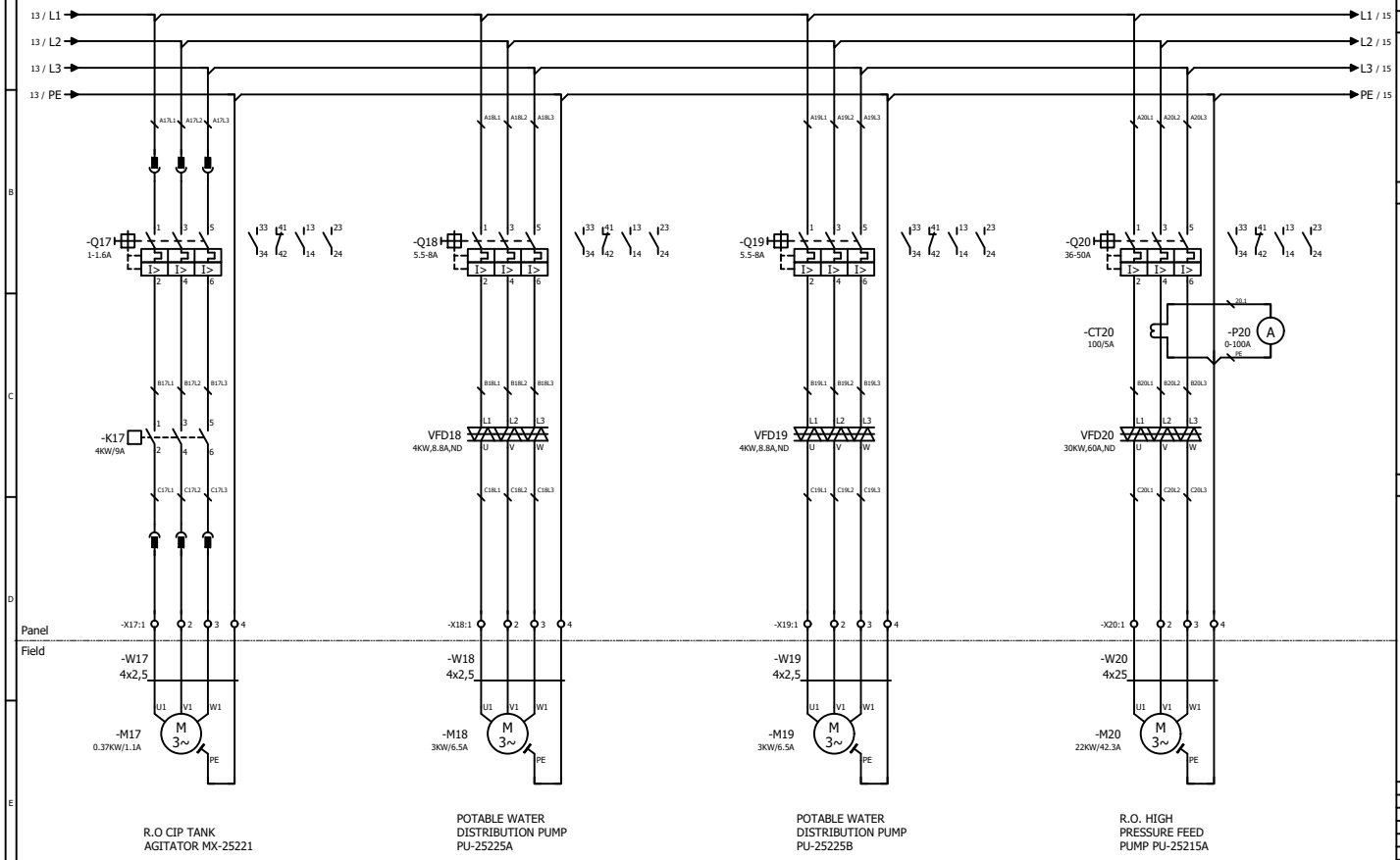
11L1-AV-P-220-EL-DIG-1210	8/25/2020	A3	NA	18 OF 35	04	04



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Schematic Wiring
Outgoing_5

Document Number:	DATE	SIZE	SCALE	SHEET NO.	REV. CLIENT	REV. VENDOR
11L1-AV-P-220-EL-DIG-1210	8/25/2020	A3	NA	14 OF 35	-	03